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Edge Computing Framework for Enhanced Robotic Adaptivity in Line-Less Mobile Assembly Systems

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Abstract

In response to the shortage of skilled workers and the increasing diversity of products and variants, integrating adaptive and flexible mobile manipulators into line-less mobile assembly systems (LMAS) stands out as a promising solution. To fully benefit from these robotic systems in LMAS, offloading computationally intensive tasks from mobile manipulators to edge computing systems is essential. This offloading of computational tasks not only lightens the processing load on individual robots but also enhances AI based perception, improving the overall system capabilities in dynamic manufacturing environments. Achieving these goals requires a robotic edge computing framework that serves as the key to flexible and adaptive mobile manipulation in LMAS. To construct such an edge computing framework, a comprehensive set of building blocks is required. These building blocks include a powerful edge system with high processing power, sensor-equipped mobile manipulators with the ability to perceive the environment, and a robust edge-to-robot connectivity. A comprehensive software stack, including software modules for localization and navigation in dynamic assembly environments, AI perception for precise pose estimation of assembly components, and motion planning for interaction between manipulators and assembly components, is crucial. Furthermore, virtualization techniques, a comprehensive deployment strategy, and a detailed description of robot hardware, site, and resources are essential. The proposed edge computing framework presents a solution that addresses mobile manipulators in LMAS and paves the way towards advanced industrial automation. Through implementation in a research assembly environment, the realized proof-of-concept showcased the feasibility of the proposed edge computing framework in a real-world pick-and-place scenario, highlighting its potential to enhance adaptivity and flexibility. There is a potential to refine and expand the framework to accommodate a broader range of industrial applications and streamline diverse manufacturing processes through the integration of mobile manipulators and edge computing within industrial settings.

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Keywords: Edge Computing; Mobile Robotics; Mobile Manipulator; Line-Less Mobile Assembly; Shared Intelligence; Flexible Assembly; Real-Time Communication

1. Introduction

The concept of line-less mobile assembly systems (LMAS) has emerged as a transformative paradigm in assembly organization, driven by the pressing challenges posed by increased product diversification and the need for reconfigurability within modern manufacturing systems [12]. In response to these challenges, LMAS seeks to mobilize resources for product assembly by either transporting resources to robots or relocating robots to resources [12]. However, the integration of mobile manipulators within LMAS introduces a suite of computationally intensive tasks that grow in complexity with the number of deployed robots. Despite the growing accessibility and affordability of on-board computing hardware, the demand for specialized hardware, such as Graphics Processing Units (GPUs) or Tensor Processing Units (TPUs), presents a challenge for exclusive on-board processing, particularly in tasks related to AI perception where the computational demands necessitate efficiency and parallel processing capabilities. Offloading computational tasks to edge computing systems provides a cost-effective solution by leveraging shared computing resources rather than equipping individual robots with standalone high-performance computing units. This paper focuses on the need for a robotic edge computing framework for LMAS by show-

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casing its potential in elevating adaptivity, flexibility and overall production capabilities in dynamic manufacturing environments. Section 2 covers the state of the art of LMAS and edge computing for mobile manipulators and derives requirements for the robotic edge computing framework for LMAS. Section 3 presents the architectural design of the proposed framework, while Section 4 provides insight into the implementation details. Section 5 serves as a summary of the paper's findings and outlines future research potentials.

2. Related Work

Mobile Manipulators in Line-Less Mobile Assembly Systems. LMAS are assembly systems that permit continuous production adaptation and optimization through individual job routes based on the entirely free allocation of jobs, goods, resources, time, and place of work within the factory. A LMAS is based on three main principles: the *clean floor* approach, the mobilization of multi-purpose resources and the highly flexible assignment of resources to jobs and jobs to locations. [12] The LMAS must demonstrate adaptivity by addressing key technological challenges, including the adaptive control of mobile robots, adaptive strategies for process control, and adaptive path planning [12]. Assembly resources in LMAS move either actively or passively [12]. Passively moving resources can be picked up by actively moving resources, such as mobile manipulators. Mobile manipulators are a key component for adaptive assembly tasks in LMAS. A mobile manipulator is defined as a robotic system that combines mobility, offered by the mobile platform, with manipulation capabilities, afforded by the mounted articulated arm, to create an unlimited manipulation workspace [24]. By enabling real-time object detection as in [25] and [27], AI-based vision algorithms empower mobile manipulators to adapt their actions and decision-making processes across a spectrum of assembly scenarios.

Edge Computing for Mobile Manipulators. Shi et al. (2016) define edge computing as the enabling technology allowing computation to be performed at the edge of the network, situated on the downstream data path of cloud services and on the upstream data path of Internet of Things (IoT) devices. The *edge* can be any computing and network resource along the path between data sources and cloud data centers. The core idea of edge computing is that edge computing should happen at the proximity of data sources. [22] Donno et al. (2019) consider three main implementations of the edge computing paradigm: *Mobile Cloud Computing* (MCC), *Cloudlet Computing* (CC) and *Mobile Edge Computing* (MEC). MCC is the concept of mobile offloading to reduce workload by outsourcing storage and processing to remote entities. CC consists of using *cloudlets* as a decentralized small-scale computing infrastructure that can be leveraged by nearby mobile devices via the Internet [21]. In MEC, the offloaded computation is situated at the edge of the mobile network, such as 5G, which provides low latency, high bandwidth and access to radio network information [6, 20]. According to Groshev et al. (2023), the system must ensure robust radio connectivity between robots and edge system

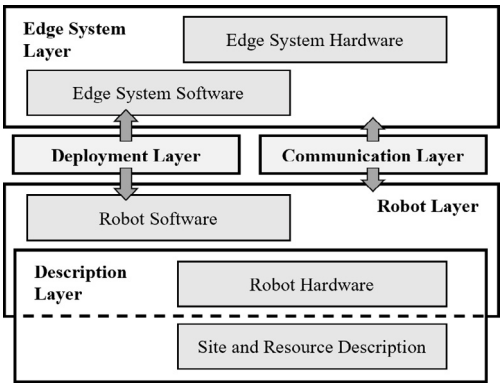


Fig. 1. Architecture of the edge computing framework

mitigating the effects of radio frequency interference. The deployment of mobile robots presents a use case for edge computing which is often related to as *edge robotics* [10]. The research of Dechouniotis et al. (2022) deals with the interplay among robots, sensors, the underlying network, and computing infrastructure. Moreover, it highlights the potential of edge robotics in facilitating resource-constrained mobile robots to offload computationally intensive tasks. [4] *KubeROS*, a platform built on Kubernetes, simplifies the deployment of robotic applications based on the *Robot Operating System 2* (ROS2) across cloud, edge, and robots and offers various deployment strategies for optimized resource utilization [28]. *RobotKube* complements this approach by introducing an event detector and an application manager to request the deployment of applications and configure the necessary changes in the Kubernetes cluster [15]. While KubeROS and RobotKube are mostly focused on the orchestration of containerized microservices, our approach centers on enhancing adaptivity and flexibility for real-world applications in LMAS, specifically targeting mobile manipulation.

Requirements for Edge Computing Framework in LMAS. The following design requirements towards an edge computing framework for mobile manipulators in LMAS can be derived from the related work introduced above: computation offloading from the robots internal processing units [6], robust connectivity between robots and edge system [10], adaptive robotic processes [12], and environmental perception capabilities [25, 27].

3. Edge Computing Framework - System Design and Architecture

The robotic edge computing framework consists of five layers, illustrated in Figure 1.: the robot layer, the edge system layer, the description layer, the communication layer, and the deployment layer.

3.1. Robot Layer

The robot layer comprises robot hardware, sensor configurations, and low-level software modules. A camera is attached

to the end effector to provide the mobile manipulator with visual perception capabilities. To further enhance those, the mobile manipulators are equipped with laser scanners that capture point clouds of the environment. Furthermore, the integration of Inertial Measurement Units (IMUs) and hall effect sensors allows for odometry calculations. Network devices are integrated into the robot units for data exchange and command transmission with the edge system. The internal computer hosts the necessary software modules that govern its low-level behaviour, such as motion execution and sensor data collection. Motion execution is controlled by robot control software that interprets higher-level commands and converts them into motor commands for both the mobile base and the robot arm. Sensor drivers enable the acquisition of sensor data, including point clouds, visual images, IMU data, and data generated by encoders.

3.2. Edge System Layer

The edge system layer serves as the central intelligence hub of the framework, responsible for sensor data processing and decision-making. The edge system comprises hardware components and software modules, empowering mobile manipulator functionalities. At its core, the Central Processing Unit (CPU) is responsible for executing a wide range of tasks, such as point cloud processing for localization and adaptive navigation. The GPU is specialized in graphical and parallel processing tailored for edge AI perception tasks, including object pose estimation. Edge AI compensates for any inaccuracies in the localization and navigation task and enables the interaction between robots and the assembly components. Random Access Memory (RAM) serves as rapid temporary storage for real-time data processing and frequently accessed information, critical for fast decision-making. Solid State Drives (SSDs) serve as data storage for extensive datasets, such as map or training data. Network Interface Cards (NICs) enable high-speed data exchange between the mobile manipulators and other devices within the network. The previously mentioned hardware components enable the execution of the following software modules.

Mapping and Localization. For robot localization, the implementation of the *cartographer* SLAM algorithm is used [11]. SLAM, as defined in [2], is a technique used in robotics to enable a robot to simultaneously map its environment and localise within it. SLAM enables rough localization of the mobile manipulator relative to assembly stations, where it performs assembly tasks.

Navigation. Navigation, as defined in [18], encompasses the autonomous movement of a mobile robot towards a specified destination, such as assembly stations, while actively avoiding collisions with obstacles encountered along its path. The proposed framework utilizes the open-source navigation stack *Nav2* [17].

Perception. The deployed perception module *DOPE* (Deep Object Pose Estimation) is a vision-based AI algorithm that analyzes camera data, with the objective of accurately estimating the pose of objects [13]. The perception module contributes to the robot's ability to localize assembly components relative to the pose of the mobile base, correcting errors stemming from imprecise localization processes.

Manipulator Motion Planning and Manipulation. Motion planning turns a high-level task command into a series of discrete motions. The utilized motion planning and manipulation module enables collision-free and precise end-effector positioning [5]. It allows for object manipulation, picking, placing, and assembly tasks based on the assembly component pose, detected by the perception module.

Finite State Machine. At a high level, the robot's behavior is governed by a finite state machine (FSM) [9]. Each state represents a specific task or behavior, such as "move to destination", "pick up object" or "perform manipulation task."

3.3. Description layer

The description layer contains description models of robots, operational site, and dynamic resources. The description of robots comprises hardware setup descriptions, specifications of the arrangement of links and joints, mesh data representations, physical properties, and robot-specific configuration files. The description of the operational site includes comprehensive map data pertaining to the environment where the robots operate, encompassing information on stationary resources, spatial boundaries, keep-out zones, and the permissible speed of robots within designated areas of the site. The description of dynamic resources contains detailed information about all resources that can be relocated during the assembly process, such as assembly components and mobile assembly stations. The description dataset of dynamic resources includes links and joints, visual and physical properties, as well as training data and trained perception models.

3.4. Communication layer

The communication layer establishes the data exchange infrastructure connecting mobile manipulators with the edge system. This infrastructure is essential for supporting the dynamic mobility of production resources across the entire shop floor. A critical requirement for this communication layer is wireless connectivity, a need that can be addressed by 5G technology. Unlike conventional bus systems, such as PROFIBUS, PROFINET, or Modbus, 5G enables communication without the constraints of physical cables, allowing for greater mobility and flexibility. Moreover, in environments with a high density of connected devices, such as an industrial setting, 5G technology outperforms traditional Wi-Fi by offering a high degree of control over the provision of wireless connectivity with strict Quality of Service requirements [19]. To ensure the robustness of the communication infrastructure, there is a need to minimize jitter, packet delay, and packet loss [14]. Jitter disrupts the synchronization of communication between mobile manipulators and the edge system while minimizing packet delay is crucial

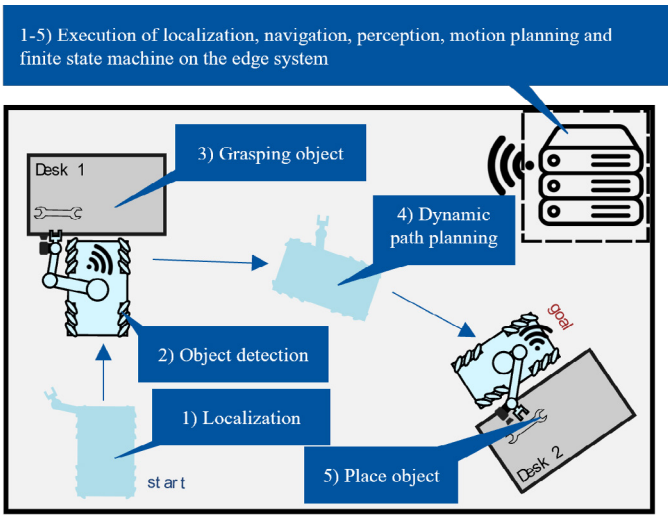


Fig. 2. Edge-accelerated mobile robot use case [1]

for achieving real-time responsiveness in time-sensitive tasks. In the context of edge robotics, the need for Ultra-Reliable Low Latency Communication (URLLC) is evident. URLLC is essential for applications requiring ultra-low latency and high reliability, ensuring that critical communication between mobile manipulators and the edge system occurs with minimal delay. Moreover, the communication layer must also address the requirements of enhanced Mobile Broadband (eMBB). As bandwidth-intensive transmission of sensor data, such as video and point cloud data, is required for robotic assembly tasks, eMBB becomes crucial for providing the necessary high data rates and increased network capacity. [7]

3.5. Deployment layer

The deployment layer is responsible for the systematic and efficient management of the transition of the software from development to deployment. It encompasses a range of tools and processes that facilitate this transition while ensuring the reliability, consistency, and scalability of the deployed software. Virtualization allows building software applications into lightweight, standalone, and executable images that contain all the necessary code, libraries, dependencies, and runtime environments. This isolation mitigates conflicts related to dependencies and libraries, enhancing compatibility, stability and hardware-agnosticism. [23] A central repository for storing and organizing these images makes them available for deployment to multiple hosts.

4. Proof of Concept

The proposed edge framework is designed to establish the foundation for the integration of mobile manipulators into the paradigm of LMAS, enhancing adaptivity and flexibility. The robotic edge computing framework has been successfully implemented and piloted, showcasing the practicality and effectiveness of the framework in various scenarios. Exemplary a pick-and-place scenario is shown in Figure 2. As mentioned in Section 3, the edge computing framework developed for mobile manipulators in LMAS can be divided into comprehensive building layers as seen in Figure 1. In the following subsections, insights are offered into each layer of the framework, showcasing the practical realization of the framework's design.

4.1. Robot layer & Edge System Layer

Hardware. The framework was tested with two identical Robotnik RB-Kairos mobile manipulators. The open-architecture computing unit running the Linux operating system enables low-effort integration into the robotics edge computing framework. The omnidirectional wheelbase and the 6 degrees of freedom (DOF) UR10 robot arm allow agile and versatile movement within confined spaces and around complex assembly setups. These robots feature a two-finger gripper, a RealSense L515 end-effector camera with RGB and depth sensors, and a pair of strategically positioned SICK S300 laser scanners for a 360° field of view. An integrated inertial measurement unit (IMU) and hall effect encoders ensure precise motion tracking. With this extensive hardware, the robots can handle various tasks, including handling, positioning, and assembly, which qualifies them for use in LMAS. According to the information presented in Section 3.2, the edge system layer serves as the central intelligence hub within the robotic edge system framework, executing the computationally intensive software components. Since the robotic software stack requires dedicated computing power, the edge system hardware components in Table 4.1 were selected for the proof-of-concept.

Table 1. Edge system hardware configuration

Component	Specification
Processors	2x AMD EPYC 75F3, 32 CPU cores, 2.95GHz
Memory	16x 16 GB RDIMM, 3.200 MT/s, Dual Rank
Storage	2x 960GB SSD SAS ISE Read Intensive, 12Gbps
Graphics Processing	NVIDIA Ampere A10 GPU, 24GB Memory

Software. The software architecture is visually represented in Figure 3 and designed to offload computation-intensive tasks from the robot's internal computation unit to the edge system. Shifting computationally intensive software components to the edge system is a strategic decision to optimize system performance and scalability within the framework. This offloading not only enhances the robot's capabilities but also ensures that the limited computational resources onboard the robots are dedicated to critical low-level control tasks. All software components within the robotic system are containerized using the container application software *Docker*, thereby ensuring the portability and isolation of their respective environments. This approach simplifies deployment procedures, enhances compatibility across platforms, and mitigates conflicts arising from dependencies and library requirements. The edge robotics framework aligns with the style of a loosely coupled software architecture,

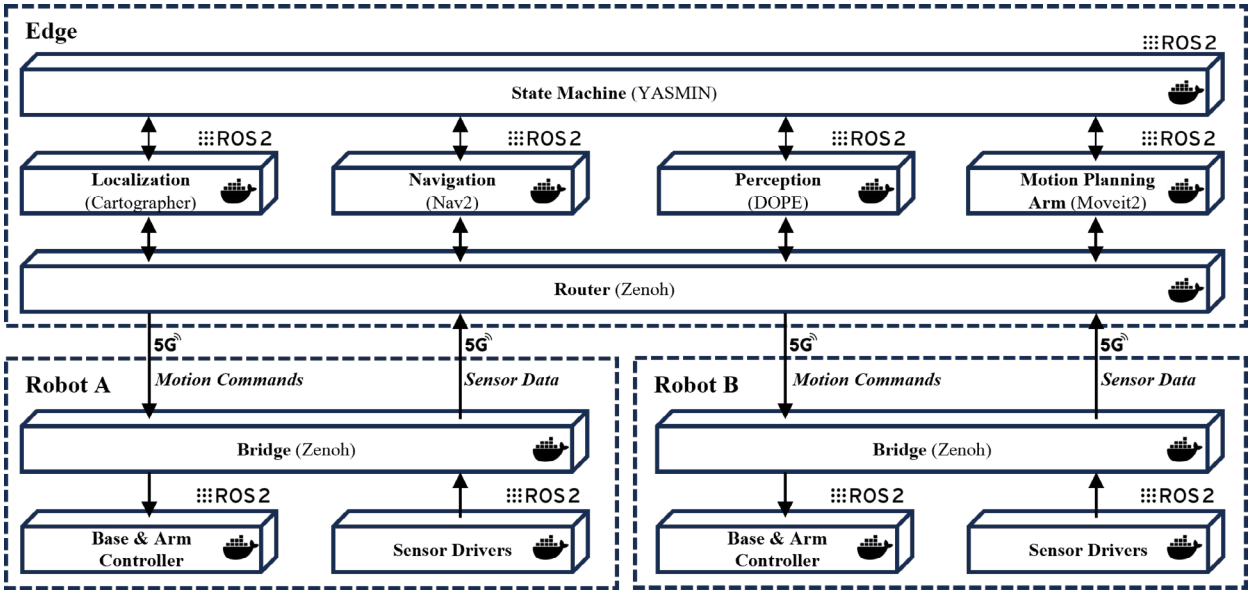


Fig. 3. Proposed edge computing software architecture [1]

emphasizing modularity, independence of components, and interoperability through well-defined interfaces. Nevertheless, the architecture has certain features that are reminiscent of a federal architecture. Notably, it places emphasis on decentralized control, allowing independent operation with a degree of self-governance.

4.2. Description layer

According to 3.3 the description layer contains the definition of the robot, the operational site, and the description of dynamic resources. The robot description encompasses details like the mobile manipulator's structure and kinematics. It defines coordinate transformations between links, such as camera and gripper, specifying how poses can be transformed from the frame of detection to the frame for grasping. Focusing on the operational site, the site description defines the maps used for localization, including goal poses for base movements, such as assembly stations, and keep-out zones, which increase safety by preventing collisions with human workers. The dynamic resource description defines the properties of objects the mobile manipulators handle during assembly tasks. It covers essential object features for grasping, such as dimensions or weight, and includes resource-specific data for AI-based pose estimation, including training images and trained models.

4.3. Deployment Layer

To address the requirements outlined in Section 3.5, a combination of Git and Docker is employed. The continuous integration and continuous delivery (CI/CD) process is initiated by pushing local Git repository changes to a remote repository, serving as a trigger for an automated workflow. The initial stage involves building a Docker image from the source code in the Git repository. In the subsequent stage, the Docker image is pushed to the edge system, rendering it prepared for execution. Depending on the CI/CD configuration of the repository, these images are then also deployed to the corresponding robots.

4.4. Communication layer

Data exchange and connectivity across the LMAS is ensured through the communication layer of the proposed framework. Strategically positioned 5G radio dots span up a 5G Radio Access Network (RAN) that covers the shop floor and connects edge system and mobile manipulators. By decentralizing computational processes and siting them close to the data source, the synergy between edge computing and 5G, as detailed in 3.4, enhances system efficiency. ROS2 is an integral element of the framework. ROS2 offers a data-centric communication model by decoupling nodes and allowing them to publish and subscribe to data topics, promoting modularity and flexibility of the framework [16]). Integrating ROS2 with the *Zero Overhead Network Protocol* (Zenoh) provides various advantages for edge robotics, where the Zenoh bridge for Data Distribution Service (DDS) aids in connecting DDS communication between ROS2 nodes. [3] Integrating ROS2 and Zenoh within the edge computing framework enhances data sharing, scalability, and efficiency, making it ideal for robotic systems in interconnected assembly environments.

5. Conclusion & Outlook

Conclusion. The proposed robotic edge computing framework is a comprehensive solution for enabling mobile manipulators to perform adaptive and flexible assembly tasks within line-less mobile assembly systems. The proposed framework consists of five essential layers which fulfill the requirements that have been derived from Section 2. The implementation in assembly scenarios, such as a pick-and-place scenario, demonstrated the

framework's practicality in enhancing the adaptivity of assembly operations. The primary strength of the robotic edge computing framework lies in its ability to elevate task execution by providing real-time edge AI capabilities and task coordination across multiple system entities. By distributing computing resources, robots are relieved of computationally intensive tasks, leaving the necessary resources available for the execution of low-level processes. Implementing the robotic edge computing framework offers increased resilience and adaptivity in assembly processes while addressing the growing need for efficiency and flexibility. As technology continues to advance, this framework provides a solid foundation for future developments in the field of mobile robotic systems.

Outlook. In future works, potential enhancements to the proposed framework for LMAS, including adaptations to handle more complex assembly tasks and the extension of the deployment layer with orchestration frameworks like *RoboCube* or *KubeROS*, will be considered [28, 15]. The integration of a digital twin layer can be explored for enabling remote robotic control, software component testing, and synthetic data generation for AI-based perception algorithms. Moreover, the potential synergy between cloud computing and edge computing can be discussed [8, 26]. Combining edge and cloud offers opportunities for real-time, low-latency processing at the edge while harnessing the expansive computational and storage capabilities of the cloud to support more sophisticated and globally connected robotic operations in LMAS. Further research considerations include addressing technical challenges, examining system requirements for 5G connectivity in industrial applications, and conducting comprehensive measurements of overall system performance.

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References

[1] Icons8, 22.09.2023. URL: https://icons8.com/license.
[2] A. R. Khairuddin, M. S. Talib, H. Haron, 2015. Review on simultaneous localization and mapping (slam), in: 2015 IEEE International Conference on Control System, Computing and Engineering (ICCSCE), pp. 85–90.
[3] Corsaro, A., Cominardi, L., Hecart, O., Baldoni, G., Enoch, J., Avital, P., Loudet, J., Guimares, C., Ilyin, M., Bannov, D., 2023. Zenoh: Unifying communication, storage and computation from the cloud to the microcontroller DSD 2023.
[4] D. Dechouniotis, D. Spatharakis, S. Papavassiliou, 2022. Edge robotics experimentation over next generation iiot testbeds, in: NOMS 2022-2022 IEEE/IFIP Network Operations and Management Symposium, pp. 1–3.
[5] D.M. Coleman, I. Sucan, Sachin Chitta, N. Correll, 2014. Reducing the barrier to entry of complex robotic software: a moveit! case study. ArXiv .
[6] de Donno, M., Tange, K., Dragoni, N., 2019. Foundations and evolution of modern computing paradigms: Cloud, iot, edge, and fog. IEEE Access 7, 150936–150948.
[7] F. Voigtländer, A. Ramadan, J. Eichinger, C. Lenz, D. Pensky, A. Knoll, 2017. 5g for robotics: Ultra-low latency control of distributed robotic systems, in: 2017 International Symposium on Computer Science and Intelligent Controls (ISCSIC), pp. 69–72.
[8] Gillani, K., Martín-Pérez, J., Groshev, M., La Oliva, A.d., Gazda, R., 2022. Don't let me down! offloading robot vfs up to the cloud, in: 2023 IEEE 9th International Conference on Network Softwarization (NetSoft), IEEE. pp. 232–236.
[9] González-Santamarta, M.Á., Rodríguez-Lera, F.J., Matellán-Olivera, V., Fernández-Llamas, C., 2023. Yasmin: Yet another state machine, Springer, Cham. pp. 528–539.
[10] Groshev, M., Baldoni, G., Cominardi, L., La Oliva, A.d., Gazda, R., 2023. Edge robotics: are we ready? an experimental evaluation of current vision and future directions. Digital Communications and Networks 9, 166–174.
[11] Hess, W., Kohler, D., Rapp, H., Andor, D., 2016. Real-time loop closure in 2d lidar slam, in: 2016 IEEE International Conference on Robotics and Automation (ICRA 2016), IEEE, Piscataway, NJ. pp. 1271–1278.
[12] Hüttemann, G., Buckhorst, A.F., Schmitt, R.H., 2019. Modelling and assessing line-less mobile assembly systems. Procedia CIRP 81, 724–729.
[13] Jonathan Tremblay, Thang To, Balakumar Sundaralingam, Yu Xiang, Stan Birchfield, 2018. Deep Object Pose Estimation for Semantic Robotic Grasping of Household Objects.
[14] Kehl, P., Ansari, J., Jafari, M.H., Becker, P., Sachs, J., König, N., Göppert, A., Schmitt, R.H., 2022. Prototype of 5g integrated with tsn for edge-controlled mobile robotics. Electronics 11, 1666.
[15] Lampe, B., Reiher, L., Zanger, L., Woopen, T., van Kempen, R., Eckstein, L., . Robotkube: Orchestrating large-scale cooperative multi-robot systems with kubernetes and ros.
[16] Macenski, S., Foote, T., Gerkey, B., Lalancette, C., Woodall, W., 2022. Robot operating system 2: Design, architecture, and uses in the wild. Science robotics 7, eabm6074.
[17] Macenski, S., Martin, F., White, R., Clavero, J.G., 2020. The marathon 2: A navigation system, in: 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), IEEE. pp. 2718–2725.
[18] Marder-Eppstein, E., Berger, E., Foote, T., Gerkey, B., Konolige, K., 2010. The office marathon: Robust navigation in an indoor office environment, in: Robotics and Automation (ICRA), 2010 IEEE International Conference on, IEEE. pp. 300–307.
[19] Oughton, E.J., Lehr, W., Katsaros, K., Selinis, I., Bubley, D., Kusuma, J., 2021. Revisiting wireless internet connectivity: 5g vs wi-fi 6. Telecommunications Policy 45, 102127.
[20] Roman, R., Lopez, J., Mambo, M., 2018. Mobile edge computing, fog et al.: A survey and analysis of security threats and challenges. Future Generation Computer Systems 78, 680–698.
[21] Satyanarayanan, M., Bahl, P., Caceres, R., Davies, N., 2009. The case for vm-based cloudlets in mobile computing. IEEE Pervasive Computing 8, 14–23.
[22] Shi, W., Cao, J., Zhang, Q., Li, Y., Xu, L., 2016. Edge computing: Vision and challenges. IEEE Internet of Things Journal 3, 637–646.
[23] Simonic, M., Pahic, R., Gaspar, T., Abdolshah, S., Haddadin, S., Catalano, M.G., Worgotter, F., Ude, A., 2021. Modular ros-based software architecture for reconfigurable, industry 4.0 compatible robotic workcells, in: 2021 20th International Conference on Advanced Robotics (ICAR), IEEE. pp. 44–51.
[24] Thakar, S., Srinivasan, S., Al-Hussaini, S., Bhatt, P.M., Rajendran, P., Jung Yoon, Y., Dhanaraj, N., Malhan, R.K., Schmid, M., Krovi, V.N., Gupta, S.K., 2023. A survey of wheeled mobile manipulation: A decision-making perspective. Journal of Mechanisms and Robotics 15.
[25] Tran, V.L., Lin, H.Y., 2021. Semantic segmentation and 6dof pose estimation using rgb-d images and deep neural networks, in: 2021 IEEE 30th International Symposium on Industrial Electronics (ISIE), IEEE. pp. 1–6.
[26] Yuan, H., Zhou, M., 2021. Profit-maximized collaborative computation offloading and resource allocation in distributed cloud and edge computing systems. IEEE Transactions on Automation Science and Engineering 18, 1277–1287.
[27] Yuan, S., Ge, Z., Yang, L., 2023. Single-camera multi-view 6dof pose estimation for robotic grasping. Frontiers in Neurorobotics 17, 1136882.
[28] Zhang, Y., Wurll, C., Hein, B., 2023. Kuberos: A unified platform for automated and scalable deployment of ros2-based multi-robot applications, in: O'Malley, M. (Ed.), ICRA 2023, IEEE, Piscataway, NJ. pp. 9097–9103.